coursera

Menu

Image Data Types

There is a wide variety of medical images. Because of this medical images are stored in a variety of ways. When working with medical images, you'll find it useful to have knowledge of the various data types used to store images so you can work with them effectively.

Navigate to the Module 1 folder in the course files and work through the **`ImageDataTypes.mlx`** script. You'll work with the X-ray image you saw in the previous video and learn how to convert between different data types.

While we recommend you run and experiment with the code in MALTAB, you can read a static version below. The interactive tools are not present in the static version.

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Introduction

All images are stored as arrays of numbers. However, because medical images are acquired with specialized equipment, the range of possible intensity values in the data often does not align with standard data types used to store variables. Being aware of the range of values in your data and the data type used to store the images will help you properly view your images.

Viewing Medical Images

Let's revisit the X-ray image of the back. It is stored as a 16-bit unsigned integer array (uint16). The intensity values for this data type range from 0 to 65,535. Small values will appear dark, and larger values bright. Try viewing the image with the `imshow` function.

Adding a colorbar can help you identify the minimum and maximum values of the data type and compare that to your data range.

```
% Load the DICOM file
backXray = medicalImage("back view01.dcm")
% extract the image
img = extractFrame(backXray,1);
maxVal = max(img, [], "all");
% view using imshow
figure
imshow(img)
title("Max Intensity = " + maxVal)
colorbar
```

The image appears completely dark because the largest intensity value in the image is 4095, or about 6% of the full intensity range.

Adjust the Display Scale

Adjust the display range to view the image. You do this using a second input argument to the `imshow` function, as described below:

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